Dr. Liam Stuart

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About

Data scientist with a PhD in mathematics and industry experience in machine learning engineering. Proven ability to translate complex mathematical concepts into actionable business insights. Experience in developing end-to-end solutions using Python and AWS to solve time series forecasting problems.

Employment

GRADUATE MACHINE LEARNING ENGINEER | SOLVEDBY.AI | SEPTEMBER 2025 – JULY 2026

Worked as an ML engineer, contributing to the full lifecycle of machine learning solutions, and leveraging fluent Python to convert research topics into production-ready code within an agile development environment.

- Actively participated in agile development workflows, contributing to code review and deployments via GitLab CI/CD pipelines, with comprehensive unit testing to ensure high code quality. Provisioned and maintained AWS cloud infrastructure using Terraform and Serverless Framework.
- Researched and implemented a multi-resolution forecasting framework that allowed users to forecast much further into the future with no impact on computational effort or performance.
- Researched and implemented a forecasting framework to automatically handle the impacts of holidays and special events, which considerably reduced manual work and improved forecast quality for users.

TEACHER OF MATHEMATICS | ELITE TUITION | SEPTEMBER 2023 – AUGUST 2025

Worked as an online mathematics tutor working with A-Level Mathematics students, aiding them in their studies and furthering their academic development.

RESEARCH FELLOW IN MATHEMATICS | UNIVERSITY OF ST ANDREWS| SEPTEMBER 2022 – MARCH 2023

Research was focused on fractal geometry and dimension theory, with a focus on hyperbolic geometry and conformal dynamics.

Education

PHD IN MATHEMATICS | UNIVERSITY OF ST ANDREWS | SEPTEMBER 2019 – SEPTEMBER 2022

Thesis title: Limit sets, Julia sets, and Sullivan's dictionary: a dimension theoretic analysis.

MMATH MATHEMATICS | UNIVERSITY OF ST ANDREWS | SEPTEMBER 2015 – JUNE 2019

Honours Average: 18.0 (Graded on a scale of 1-20).

Other

SCIENCE TO DATA SCIENCE BOOTCAMP (S2DS) | MARCH 2025 – APRIL 2025

Worked with a team to develop a proof-of-concept end-to-end pipeline for detecting counterfeit designer handbags, with the goal of aiding potential stakeholders such as law enforcement and retailers of luxury fashion items. Main tools included web scraping for data collection, NLP and sentiment analysis for feature engineering, and clustering for classification.

Skills & Abilities

Languages & ML – Python, SQL, Time Series Forecasting, Clustering, Gradient Boosting, Scikit-Learn, TensorFlow, PyTorch.

Cloud & DevOps – AWS (API Gateway, DynamoDB, ECR, Lambda, S3), Docker, Terraform, Serverless Framework, Git, GitLab CI/CD, REST APIs (OpenAPI/JSON Schema), Agile.

Certifications

DEEP LEARNING/MACHINE LEARNING SPECIALISATIONS | DEEPLEARNING.AI (VIA COURSERA) | 2024

IBM DATA SCIENCE PROFESSIONAL CERTIFICATE | IBM (VIA COURSERA) | 2024

Hobbies & Interests

Playing pool, stationary cycling, reading manga, playing video games, speedcubing.